2^{nd} Homework

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I. Introduction

The purpose of this homework, is to work on facial image datasets, using Principal Component Analysis, then train a classifier to recognize Bruce Lee face. This document will present an overview of every major section in the homework, but it will not go in detail about every subsection, or explicitly mention lines of code; instead it'll focus on more on the results obtained.

The dataset given for the exercise will be passed trough transformations, with the purpose of creating more datasets. This is done to see what transformation provides less overlap between classes, or what dataset has more variance. In turn, the previous information will help decide which dataset is the best to train the classifier.

II. Data loading

The dataset for the exercise contains faces of Bruce Lee, Neil Patrick Harris and Pam Grier.

Pre-processing of the dataset is made using the tool MCTNN, which helps with detecting faces of the images and aligning them. With this, the faces are resized into images of 256 x 256 pixels. This faces are then put into a tensor.

From now on, four datasets are created:

The first dataset is the unaltered output from the pre-processing. [Figure 1]

Figure 1: Dataset with no filters

From the previous one, a second dataset is created, where the images are converted into grayscale. [Figure 2]

Figure 2: Grayscale Dataset

The third dataset is the output of the first dataset, after passing trough a binary black and white filter, with the threshold at the half scale of brightness. [Figure 3]

Figure 3: Binary Black and White Dataset

Similarly, the fourth dataset is the first dataset, passed trough a gaussian blur filter; with a kernel size of 15, and a variation between 3 and 7. [Figure 4]

Figure 4: Gaussian Blur Dataset

Having the four datasets, the next step is to perform Principal Component Analysis.

III. Data projection

Principal Component Analysis (Or PCA for short) is a method used to reduce multidimensional datasets into a lower dimensional space (in this case, 2D), to ease analysis of overlap between classes, or variance.

Principal Component Analysis (PCA) was applied to the four datasets obtained after preprocessing the original RGB images, grayscale, black-and-white, and Gaussian-blurred versions. The objective was to project the high-dimensional facial data into a two-dimensional space, allowing visualization of class separability and an understanding of how each transformation affects the intrinsic structure of the data.

For this section, the dataset is transformed into linearly independent data, by associating its variance to a vector that best fits a linear regression (Principal components). This allows to reduce dimensionality without significant loss of information.

After obtaining the first two principal components of each dataset, the data of each is projected on a scatter plot; where each dot is an image, and each color is who the image represents. To this scatter plot, another graph is added below, which represents how the number of principal components affects the cumulative variance of the dataset.

The graphs are shown below:

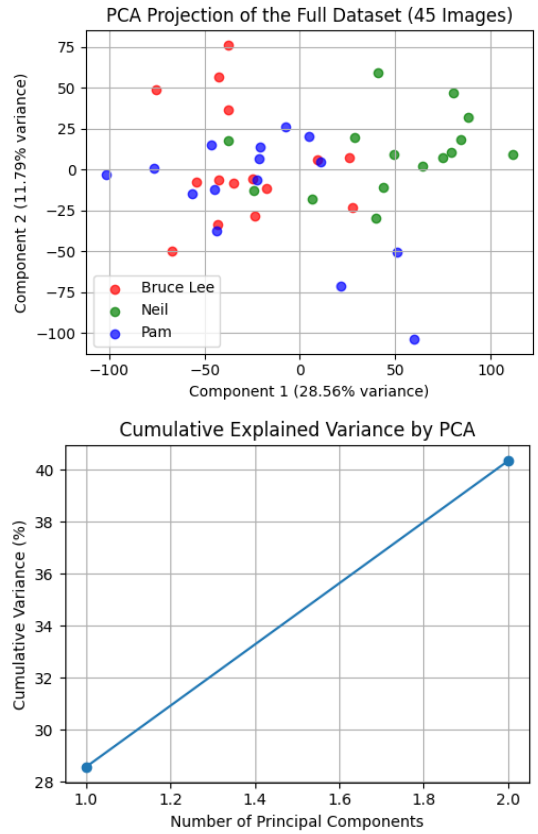

Figure 5: No filter Dataset PCA

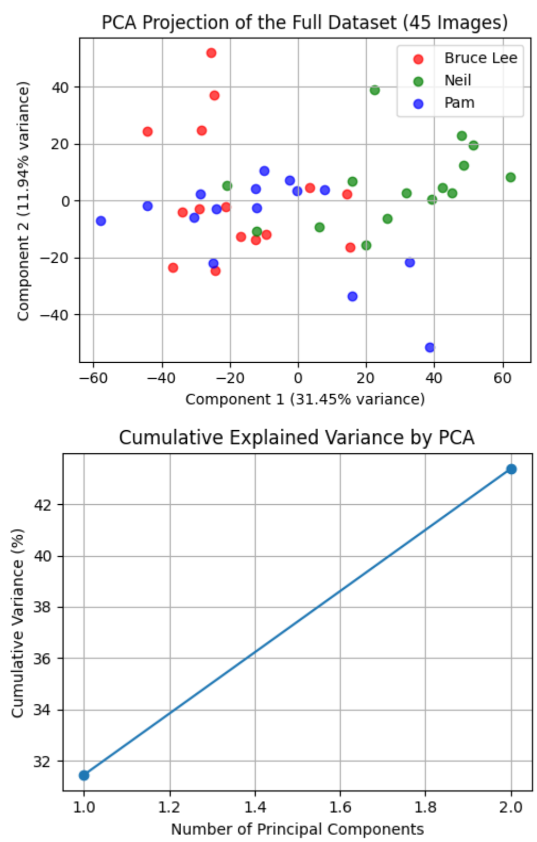

Figure 6: Grayscale Dataset PCA

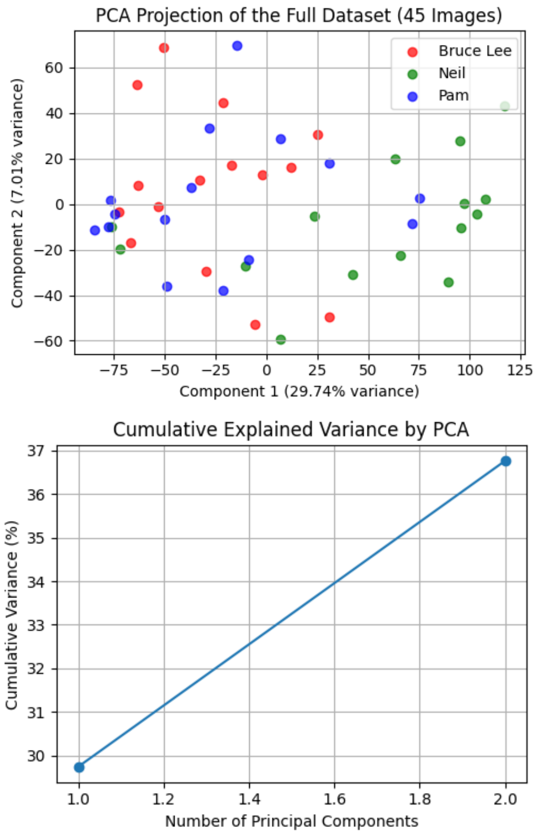

Figure 7: Binary Black and White Dataset PCA

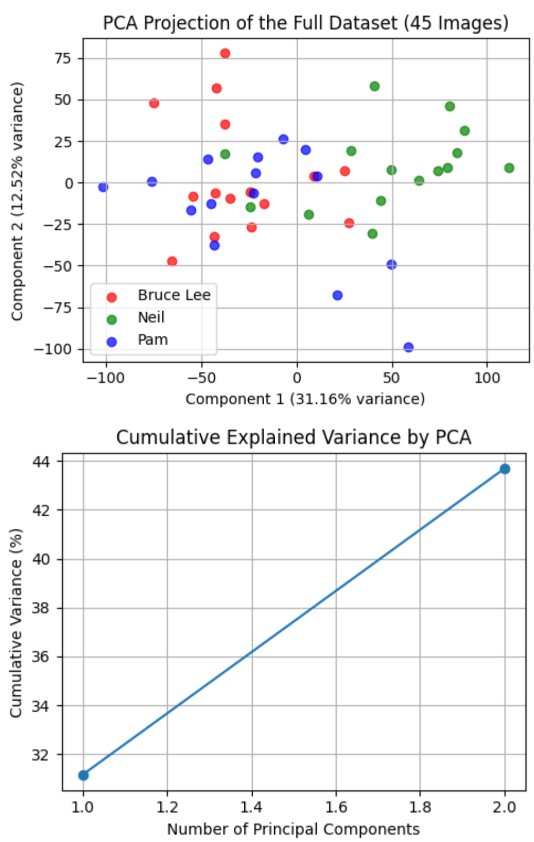

Figure 8: gaussian blur Dataset PCA

The Explained variance in the original [Figure 5] and gaussian blur dataset [Figure 8] are roughly the same, with about 40% on cumulative variance for two Principal components. The grayscale dataset [Figure 6] has slightly more cumulative variance that the two previous datasets. And the binary black and white dataset [Figure 7] has the lowest variance; at around 37%. From the previous information it's possible to extrapolate that the grayscale dataset does a better work at retaining more representative information at two Principal Components, but it does not provide information about clustering of the different classes.

Returning to the previous thought, while classes are not the most defined; and present a good amount of overlapping, the original dataset presented a more clear distinction between groups. The gaussian blur dataset showed similar clustering to the original dataset; and the binary black and white dataset presented some good spacing between classes. The grayscale dataset on the other hand showed a substantial amount of overlapping; meaning that in the transformation, facial features were lost and found irrelevant within the first two Principal Components.

Overall, the PCA analysis revealed a trade-off between compactness and separability. While some transformations increased statistical variance retention, they also reduced the distinctiveness of features crucial for classification. Therefore, with consideration of all conclusions reached above, the dataset that is considered best suited for the training of a classifier model, is the original dataset.

IV. Classifier Model

To evaluate the effect of data representation on model performance, a binary logistic regression classifier was trained using three different datasets: the original RGB images, PCA-reconstructed images, and noise-corrupted images. The classifiers goal was to distinguish Bruce Lees face from the other two individuals, providing a practical scenario to analyze how preprocessing choices influence bias, variance, and generalization capability.

The logistic regression model was chosen due to its interpretability and linear decision boundary, which makes it sensitive to changes in data distribution and feature quality. When trained on the PCA-reconstructed dataset, the model behaved as if it were regularized: the dimensionality reduction simplified the input space, effectively filtering out noise and redundant information. This reduction in complexity lowered the variance, as the model became less sensitive to small fluctuations in the data. However, the same simplification introduced a noticeable increase in bias the reconstructed images lacked fine facial details such as contour edges and texture gradients, leading to a systematic underfitting of the real data patterns.

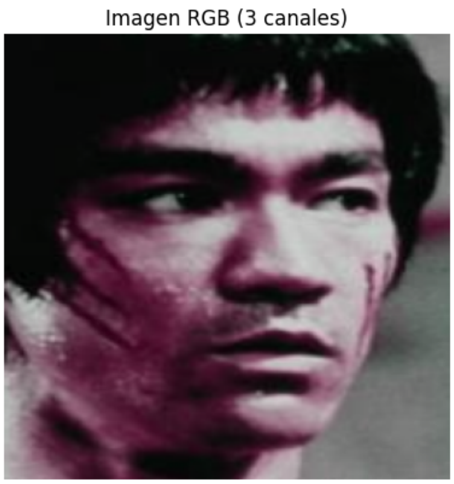

Figure 9: Original Dataset training image example

In contrast, training with the noise-corrupted dataset produced the opposite behavior. The addition of Gaussian noise introduced random perturbations that distorted the underlying distribution, forcing the classifier to adapt to irrelevant variations rather than meaningful facial features. This increased the variance substantially, as the models predictions became more unstable and dependent on the random noise present in each sample. Furthermore, high noise levels also increased bias slightly, since the models capacity to capture the true underlying relationships was compromised by the corrupted signal.

Empirically, the accuracy decreased in the noisy dataset compared to both the original and PCA-reconstructed versions. The resulting decision boundaries were less defined, confirming that noise increases the models uncertainty and reduces class separability. The clean dataset, by contrast, provided the optimal balance between bias and variance, preserving essential discriminative features while maintaining signal consistency across samples.

Figure 10: Noise Dataset training image example

These results reinforce the theoretical biasvariance trade-off; while PCA acts as a regularizer that simplifies the model and reduces overfitting, excessive simplification or data corruption both degrade generalization. The experiment demonstrates that data quality and representation directly determine the classifiers stability and predictive performance, highlighting the importance of preprocessing decisions in the design of robust machine learning pipelines.

Figure 11: Gaussian Blur image example for test the trained model obtaining a high accuracy

V. Conclusion

The implementation of image preprocessing, dimensionality reduction, and classification techniques allowed us to analyze how data transformations affect model performance and generalization. Through PCA, we observed that although grayscale images captured slightly more cumulative variance, the original RGB dataset preserved richer discriminative information, resulting in clearer class separation. Excessive simplification, such as binarization or blur filtering, caused significant overlap between clusters, demonstrating the importance of detailed visual features for effective representation.

The logistic regression classifier confirmed these findings. Training on PCA-reconstructed data reduced variance but increased bias, while adding Gaussian noise led to higher bias and variance simultaneously, degrading performance and decision boundary clarity. These results highlight the delicate trade-off between data simplification and information preservation.

We could conclude that the original dataset offered the optimal balance of bias, variance, and signal quality. Maintaining the richness of the input data proved essential for both dimensionality reduction and classification accuracy. This exercise emphasized how preprocessing choices directly influence model stability, interpretability, and overall robustness in facial recognition tasks.